

SOVEREIGN GAME

A Strategic Simulation Environment for Deep RL

Project Presentation

CS272 Reinforcement Learning and Sequential Decision Making

Garvit Sharma and Yukta Vajpayee

GitHub Repo:

<https://github.com/GarvitSharmaDeveloper/SOVEREIGN-GAME>

YouTube Link:

<https://youtu.be/yyn9iT0553E?si=2YgOCrCEUCEmgalC>

CAN A MILITARILY SUPERIOR AGENT LEARN THAT INVASION IS STRATEGICALLY DOMINATED?

Investigating if an AI can deduce that peace is the optimal strategy through experience alone in a stochastic-politics simulation.

THE GAME

A deterministic-core, stochastic-politics simulation involving exactly **three nations**.

INVADER (AGENT)

Starts with a massive hard-power advantage (**2x units**). Goal: Maximize long-term reward.

DEFENDER

Rule-based opponent with a **home-turf advantage**. Defends territory against aggression.

NEUTRAL

Stochastic observer whose **political posture** shifts based on the Invader's actions.

CUSTOM GYMNASIUM ENVIRONMENT BUILT FOR COMPLEX GEOPOLITICAL STRATEGY

FRAMEWORK

SOVEREIGN ENV

Built from scratch as a custom **Gymnasium** environment, ensuring full control over state transitions and reward logic.

ALGORITHM

PPO

Proximal Policy Optimization via Stable-Baselines3. Chosen for its stability and efficiency in high-dimensional spaces.

STATE SPACE

MULTI-FACTOR

- 9-territory map status
- Military strength
- International legitimacy
- Supply index
- Neutral posture

ACTION SPACE

MULTIDISCRETE

Allows the AI to take one **Political** action and one **Military** action simultaneously.

RL DESIGN CHOICES: DENSE REWARDS AND DRIFT-DIFFUSION MODELS ACCELERATE AGENT CONVERGENCE

DENSE REWARD FUNCTION

- **Continuous Feedback:** Points and penalties are awarded every step rather than just at game end.
- **War Costs:** Simulates the immediate financial and political drag of military occupation.
- **Faster Convergence:** Provides a clearer gradient for the PPO agent to optimize its policy.

DRIFT-DIFFUSION MODEL

- **Stochastic Politics:** Combines deterministic reactions with stochastic noise to mimic real-world unpredictability.
- **Neutral Posture:** Governs how the international community shifts from observer to active sanctioner.
- **Mathematical Rigor:** Ensures the environment is not just a game, but a robust geopolitical simulation.

RL DESIGN CHOICES: SYSTEMATIC ABLATION STUDIES EVALUATE THE IMPACT OF POLITICAL CONSTRAINTS

METHODOLOGY

Systematically turning off (ablating) specific environment rules to isolate their impact on the Agent's emergent policy and decision-making.

Parameter Tuning:

Tested "Early Sanctions" to analyze how lowering the threshold impacts AI behavior.

THE 5 CORE EXPERIMENTS

1. Full Model

All political, economic, and international rules active.

2. No Legitimacy Penalty

Removing the cost of international reputation.

3. No Occupation Cost

Eliminating the economic drag of holding territory.

4. No Neutral Posture

International community ignores the war.

5. Baseline (Arcade Mode)

All rules off, focusing purely on military conquest.

RESULTS: PEACE EMERGES AS THE MATHEMATICALLY OPTIMAL STRATEGY IN THE FULL MODEL

~40.26

HIGHEST MEAN REWARD ACHIEVED

100% Frequency of the
NEGOTIATE Action

THE OUTCOME

In the "Full Model", a **Peace-Dominant Policy** definitively emerges as the agent's primary strategy.

STRATEGIC DEDUCTION

The AI mathematically deduced that the compounding, long-term costs of war, **economic sanctions** and **legitimacy collapse** which completely outweigh short-term territorial gains.

DIPLOMATIC RESOLUTION

The game consistently concludes via **Diplomatic Resolution**, proving that the agent prioritizes sustainability over raw conquest.

RESULTS: POLITICAL CONSEQUENCES DRIVE THE SHIFT FROM CONQUEST TO DIPLOMACY

BASELINE CONTRAST

0%

Negotiation frequency in the Baseline model. Without political costs, the AI realizes diplomacy is **mathematically impossible** and defaults to pure conquest.

EARLY SANCTIONS

FAST

Lowering the sanction threshold causes the AI to seek diplomatic resolutions **even quicker** to avoid accelerated economic drag and resource depletion.

THE "STRIKE" ACTION

0%

The AI optimized STRIKE out of its playbook. **ADVANCE** is politically cheaper, preserves finite units, and provides immediate resource points.

REALISTIC MULTI-DIMENSIONAL COSTS FOSTER ORGANIC PURSUIT OF PEACE

PROJECT SUMMARY

SOVEREIGN proves that a militarily superior PPO agent will **organically pursue peace** if an environment enforces realistic economic, political, and international costs.

FUTURE IMPACT

This research establishes a strong foundation for utilizing **Deep Reinforcement Learning** in geopolitical forecasting and conflict resolution studies.

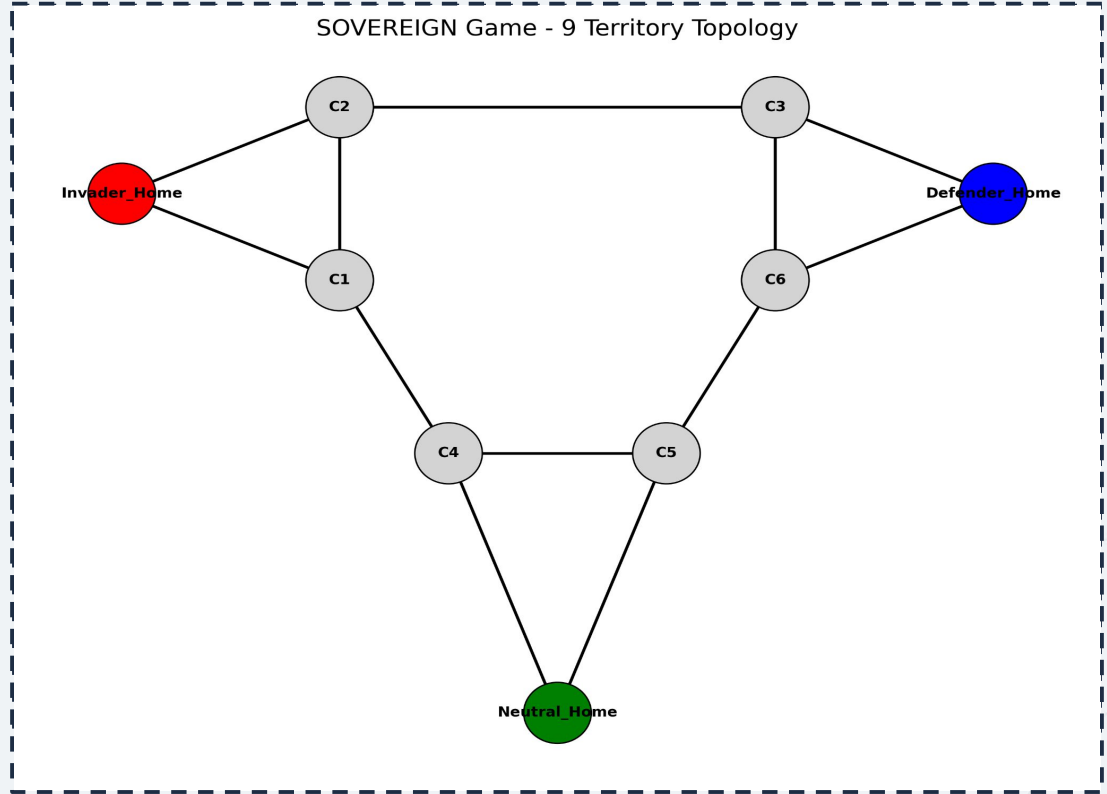
*"Strategic dominance is not just about military power, but about the **long-term sustainability** of the resulting geopolitical state."*

APPENDIX: 9-TERRITORY MAP TOPOLOGY

NOTE FOR PROFESSOR

This 9-node graph illustrates the connections between the **Invader Home**, **Defender Home**, **Neutral Home**, and the 6 Contested Territories.

Supply lines are mathematically tied to this topology; non-contiguous occupied territories generate **no resource value** and increase occupation drag.



APPENDIX: EXPERIMENTAL ACTION DISTRIBUTION

NOTE FOR PROFESSOR:

"This chart visualizes the stark contrast in the AI's action frequencies. Notice the heavy reliance on 'NEGOTIATE' (green) in the Full Model versus the absolute dominance of 'ADVANCE' (orange) in the Baseline where political consequences were ablated. 'STRIKE' (red) remained globally suboptimal."

● NEGOTIATE ● ADVANCE ● STRIKE

